ChE 382: Unit Operations Laboratory

ULTRAFILTRATION
Revision: 2019v1

Introduction

The purpose of this laboratory experiment is to examine an ultrafiltration system and its filtering effectiveness. Ultrafiltration is a process that can be used to concentrate solutions containing small macromolecules or colloidal sized particles. For basic ultrafiltration, the solution is pumped across a membrane that is permeable to the solvent but that completely retains all solutes of molecular weight above the molecular weight cutoff of the membrane.

Ultrafiltration is a relatively new unit operation in chemical engineering. It emerged in the 1970s as polymer chemists and engineers discovered techniques to fabricate porous polymeric membranes which can separate on a molecular scale. It can be considered as molecular filtration. It is presently used in the purification of natural products, the recovery of pharmaceuticals, and has applications for environmental management.

Ultrafiltration can be utilized to accomplish one or more of the following separations: (1) the concentration of solute by removing solvent from solution; (2) the purification of solvent by removing solute from solution; or (3) the separation of Solute A from Solute B by using a membrane permeable to A only (or vice versa). In this experiment, the first process mentioned is performed in which a milk-water mixture is filtered to concentrate the milk (solute).

An ultrafiltration membrane is typically manufactured from a thin porous plastic film. The pores in the membrane film are typically 10-30 nanometers in diameter. These pores are large enough to allow the solvent molecules to pass through, but they are too small to allow larger macromolecules to pass. The ultrafilter used in this experiment is a hollow fiber membrane. The filter element consists of a bundle of small hollow tubes. The solution to be filtered flows inside of the hollow fibers; the solvent molecules, which can pass through the pores flow radially out through the wall of the hollow fiber.

In selecting ultrafiltration to recover solvent from a solution, consideration needs to be given to other unit operations such as evaporation, which can be used for solvent recovery. The book by King [1] gives a good description of ultrafiltration and also a discussion of the criteria used to select the optimal separation process for a given application.

Ultrafiltration is in a different class of process than distillation and solvent extraction. It depends upon the relative rate of permeation of components through the membrane while distillation and extraction depend upon equilibrium between immiscible phases.

Theory (See Refs [1] and [2] for more details)

The following mass balance can be applied to the ultrafiltration system:

$$Mass\ feed = Mass\ concentrate + Mass\ permeate$$

Here the concentrate is the macromolecular solutes, colloids, dispersed phase, or other species that are too large to pass through the membrane surface, and the permeate is the fluid or solution that passes through the membrane surface.

In this experiment, the volumetric flowrates for an ultrafiltration process are measured at several pressure differentials. Several runs are made at different pressure differentials in which a percentage of the solution is removed. A graph of the Concentration Factor (CF) versus the flux is plotted to determine the optimum operating pressure for the filter.

The Concentration Factor (CF) is defined as:

$$CF = \frac{V_{original}}{V_{concentrate}} = \frac{V_{original}}{V_{original} - V_{permeate}}$$

Concentration Factor can also be defined as:

$$CF = \frac{1}{1 - R}$$

Where R is the recovery and is defined as the feed stream that becomes permeate:

$$R = \frac{V_{permeate}}{V_{original}}$$

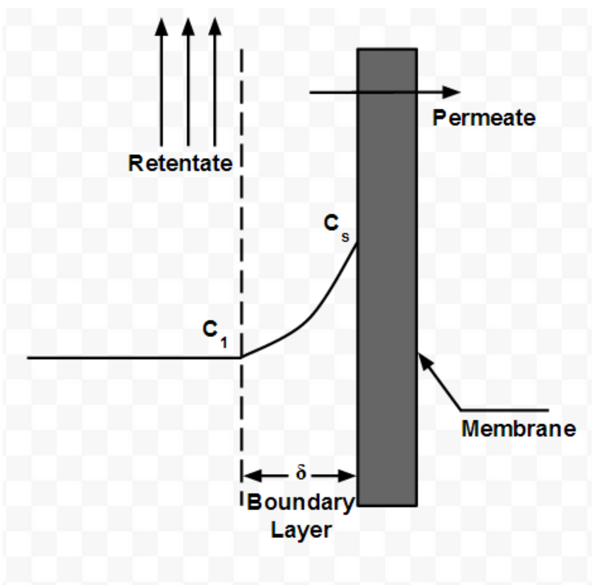
Permeate flux is measured for various recoveries and differential pressures. Flux is the rate of permeate flowing through a unit area of the membrane surface, A plot of permeate flux versus concentration factor allows for the determination of pressure differential where the greatest flux is obtained.

The basic model of ultrafiltration expresses the solvent flux through the membrane as a function of applied pressure:

$$\nu = Q_{\nu \Delta P} \tag{1}$$

Where Q_υ is the membrane permeability for the solvent and υ is the volume flux. This expression assumes that the solution is sufficiently dilute that osmotic pressure effects do not influence the driving force. It also assumes that the solutes do not permeate through the membrane.

The non-permeating solute does, however, concentrate at the membrane surface. This is called concentration polarization. It is shown schematically below.



At steady state, the convective flow of solute toward the surface must equal its back solution into the bulk. This mass balance over a thin element near the surface gives:

$$C_1 \nu = -D_\nu \frac{dC_1}{dx} \tag{2}$$

where D_v is the solute diffusion coefficient and C_1 the solute concentration in the bulk of the solution. Subject to the two boundary conditions, the solution of the differential equation gives the flux as

$$\nu = k_c \ln\left(\frac{C_s}{C_1}\right) \tag{3}$$

Where

C_s = Solute concentration at the surface

k_c = mass transfer coefficient $\left(\frac{D_\nu}{\delta}\right)$

δ = thickness of concentration boundary layer

D_υ = Diffusion coefficient

It is surprising that the flux does not depend on pressure. Experimental studies show that the flux quickly reaches a constant value and does not increase as the applied pressure is increased. In the early years of ultrafiltration this confused many researchers. It took chemical engineering analysis to show that the flux depends upon a balance of convection and diffusion in the boundary layer. See reference [3] for more details.

Equation 3 shows that the solvent flux versus the logarithm of the feed concentration should be a straight line if the surface concentration is constant (this is the case at the solubility limit).

Equation 3 provides a simple test for when concentration polarization is important in the ultrafiltration process. If C_5 is small, then:

$$\frac{k_c}{\nu} \gg 1$$

In many mass transfer processes, the boundary layer can be estimated to be in the range of 0.01 cm.

Experimental Unit

The experimental unit consists of a hollow fiber module connected to tanks for the process feed and the permeate product. The feed pump delivers a constant head of pressure and valves on the inlet and outlet to the module are adjusted to fix the inlet and outlet pressure. A schematic of the ultrafiltration process is shown in Figure 1.

An important issue with this filter is that it has to be cleaned before starting an experiment. Then, a water flux test is run to check whether or not the filter is clean. Only after the cleaning cycle is completed can the experiment be initiated.

Coordinating Cleaning Cycles and Experiments between Lab Teams

The cleaning procedure is fairly long with six cycles and requires almost the full pre-lab period to perform. It will be split into two parts on the prelab week, with the morning (AM) lab team performing procedures (1) to (3), and the afternoon (PM) lab team performing (3) (repeat) to (5). **Cleaning step (6), the sanitation cycle, is not done until the final lab period the next week by the morning (AM) team.**

When your team (both AM or PM periods) completes the final lab period experiments, perform a drain and rinse cycle (3) so that the column is ready for the next team. The afternoon (PM) teams **do not** have to redo the sanitation cycle that was done in the morning (AM).

Cleaning Procedure

(1) LIQUID SOLUTION FLUSH

A. Fill the process tank with 20 L deionized water at approximately 105°F. Open the Intake-Pump valve (PV1).

B. Direct the permeate outlet flow to the process tank

C. Close the inlet pressure control valve (V1) and open the outlet pressure control valve (V2)

D. Start the circulation pump

E. Slowly open the inlet pressure control valve (V1) until a pressure of 20 psi is obtained on the adjacent pressure gauge (P1).

F. Slowly close the outlet pressure control valve (V2) until a pressure of 8 to 10 psi is obtained on the adjacent pressure gauge (P2).

G. Readjust the inlet pressure control valve (V1) until the inlet pressure is stabilized at 25 psi and the outlet pressure is stabilized at 10 psi.

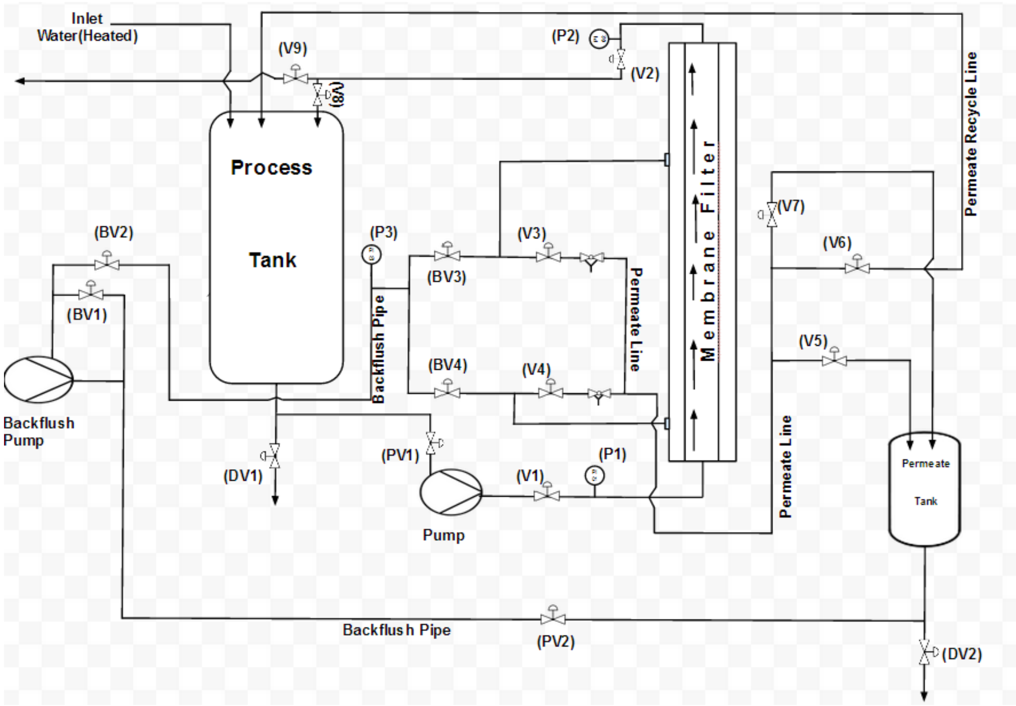


Figure 1. Schematic of Ultrafiltration System

H. Allow the system to run for 10-15 minutes. During this time check the system for leaks. There should be NO AIR BUBBLES visible in the tank. If air bubbles are present, check with an instructor.

I. Slowly open the outlet pressure control valve (V2) fully.

J. Close down the inlet pressure control valve (V1).

K. Stop the circulation pump.

L. Drain the water from the process tank and filter.

M. Rinse the process tank.

(2) ACID CYCLE

A. Make up 20 L of approximately pH 2.00 phosphoric acid solution in deionized water at 105°F. ($K_{P1} = 7.11 \times 10^{-3}$)

B. Repeat steps B to G of the Liquid Solution Flush

H. Run in this process mode for 10 minutes. Next close off the permeate flow valves (V3) (V4) and run the acid solution through the filter for another 5 minutes.

I. Repeat steps I to M of the Liquid Solution Flush

(3) DRAIN AND RINSE CYCLE

A. Refill the process tank with 50 liters if 105°F deionized water.

B. Direct the permeate outlet flow to permeate tank.

C. Open Permeate Drain Valve (DV2).

D. Repeat steps C to G of the Liquid Solution Flush. Momentarily crack the drain valves during the rinse cycle to flush out any trapped acid solution. *Make sure to stop the pump before the water level in the process tank falls below the level of the inlet pipe to the pump.*

H. Rinse Permeate Tank

(4) CAUSTIC CYCLE (Save Permeate)

A. Make up 20 L of a 1% sodium hydroxide solution in deionized water at 105°F. Use weight basis (density of water is 8.33 lbs/gallon).

B. Repeat steps B to M of the Liquid Solution Flush, except direct the permeate to the permeate tank instead of the process tank. Make sure the drain valve on the permeate tank is closed. The permeate solution is to be saved for backflushing.

(5) BACK FLUSH MODE

A. For the permeate area, close the permeate recycle valve (V6). Open the permeate pump valve (PV2). Close the permeate tank drain valve (DV2). Open the permeate drain valve (V5). In the back flush area, close the recycle back flush valve (BV1) and open the back flush return to system valve (BV2). In the front, open the lower left back flush flow valve (BV4), the upper left backflush flow valve (BV3) and output process valve (V2). All other valves in the front should be fully closed except Intake Pump Valve (PV1).

B. Turn on back flush pump

C. Allow the permeate liquid to flow from permeate tank, through the system via back flush, and into the process tank)

D. Once the permeate tank level halves, Close the process tank drain valve (DV1)

E. Drain the permeate tank, close all back flush valves, and open outlet pressure valve

F. Turn off back flush pump, turn on circulation pump

G. Slowly open the inlet pressure valve to 20 psi

H. Slowly close the outlet pressure control valve to 10 psi

I. Adjust the valves until the inlet is set to 25 psi and outlet is set to 10 psi

J. Run the system for 15 minutes

K. Check for any leaks in the system or air bubbles in the tank

L. Slowly open the outlet pressure valve and close the inlet pressure valve

M. Drain the filter and the process tank

N. Rinse the process tank to prepare for the next cycle

(6) SANITATION CYCLE

A. Make up a 200 ppm (200 ppm available chlorine) sodium hypochlorite solution in 20 L of deionized water at 105°F. (Use 350 mL household bleach/50 L water.)

B. Repeat steps B to M of the Liquid Solution Flush

Repeat the drain and rinse cycle (3) described above.

Experimental Measurements

The objective of the experiment is to measure the flux through the membrane for different pressure drops and for different milk concentrations (i.e., recoveries). Different milk concentrations are obtained as follows: The feed is run through the membrane until the appropriate amount of solvent (permeate) is removed. At this point, the permeate is directed back to the feed tank and the flux measurement is made. When the permeate flow is directed back to the feed tank, no further recovery is obtained and the feed concentration does not change with time. This allows one to measure the permeate flux at constant feed concentration.

* The pressure in hollow fiber tubes may be taken to be the average of the pressure at the inlet and outlet of the filter.

Experimental Procedure

WATER FLUX TEST

After the cleaning procedure has been completed, determine the permeate water flux of the ultrafilter at an inlet pressure of 25 psi and outlet pressures of 5, 10, 20, and 15 psi using 20L of 105°F deionized water . The flux should be measured in a suitable graduated cylinder to allow a time lapse of at least 30 seconds.

MILK TEST (Whole, 2%, 1% or Skim milk will be used)

After the Water Flux Test

A. Fill the process tank with 80 liters of 105°F deionized water. Add 1 gallon of milk and stir.
   NOTE: DO NOT START THE SYSTEM UNTIL THE FEED MATERIAL IS AT THE CORRECT TEMPERATURE. LOW TEMPERATURE FEED MATERIAL CONTACT CAN CAUSE PREMATURE FOULING OF THE MEMBRANE SURFACE AND BIAS THE DATA COLLECTED. IT IS WORTH THE EXTRA TIME TO STABILIZE THE FEED TEMPERATURE.

Start the system in the normal manner and adjust the pressure to 25 psi inlet and 5 psi outlet. Run in this manner until 10% recovery is reached. For example, if 100 liters of feed material is used at 10% recovery, there would have been 10 liters of permeate removed:

$$\%\ Recovery = \frac{Volume\ Permeate}{Volume\ Original} \times 100\%$$

Return the permeate line to the process tank and maintain this operation mode (no recovery) while performing the permeate flux measurements. Under this mode of operation no concentrating is being done and the feed concentration remains approximately constant. (Obviously some solvent is removed during the measurement itself, but this can be returned to the process tank after the measurement is completed.)

The permeate rate is related to the combined effect of flow and pressure through the hollow fibers. The flow rate is changed by increasing or decreasing the process outlet pressure (V2 read at P2) while maintaining the fixed 25 psi inlet pressure to the cartridge. When changing from one pressure to the next, wait two minutes before taking a reading. It is recommended that the process outlet pressure is varied in the following order: 5, 15, 10, 20, and 5 psi.

Repeat the experiments at recoveries of 25% and 40%.

Data Analysis

The equations developed in the Theory section suggest logical plots to interpret the data.
The data analysis should focus on identifying the effect of the process variables which can be controlled on the flux through the membrane. For example, plot the permeate flux as a function of the outlet pressure (or average pressure drop across the membrane) for the Water Flux Test and for the various milk concentrations (recoveries) at which flux measurements were obtained.

What were the optimum pressure settings for the greatest flux? Did you see any effect of concentration polarization during your experiments?

References

[1] C. J. King, *Separation Processes*, 2nd ed. McGraw-Hill, 1980.

[2] W. L. McCabe, JC. Smith, P. Harriott, "Unit Operations of Chemical Engineering", pp.1037 – 1058, 7th ed. McGraw-Hill, 2005.

[3] Porter, Mark, "Concentration Polarization with Membrane Ultrafiltration", Ind. Eng. Chem. Prod. Res. Dev. Vol. 11, No. 3, pp. 234-248, 1972.